

Week 1 Project Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Course Name: ECE501 Digital Image Processing (DIP)

Group Number: Group 15

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Week Number: Week 1

Date: September 13, 2026

Objectives of the Week

- Finish defining the project topic, scope, and problem statement.
- Be aware of the real-world challenges of defect detection on fabrics by image processing methods.
- Create the repository for the project and establish a basic plan for the coming weeks.
- Perform a prior literature survey on defect detection on fabrics and classical texture features.

Work Completed This Week

- Confirmed the project topic in accordance with the course guidelines for the group project.
- Identified the objectives and scope that are expected from the project.
- Created the repository for the project and organised the basic folder structure for the project (data, notebooks, src, docs).
- Discussed roles and established a workflow for tracking progress on a weekly basis.
- Researched on fabric surface defect detection techniques.
- Identified the different types of fabric defects, such as holes, stains, broken yarns, misweaves, etc., to identify what needs to be detected by the system.
- Studied classical texture features (GLCM, LBP, and Gabor Filters) at a high level of abstraction.
- Begun looking for possible public datasets for fabric defect detection (such as AITEX/TILDA style dataset).

Literature Review

Topic/Paper	Key Takeaway	Relevance to the Project
Fabric defect detection techniques ¹	provides a general understanding of techniques used to detect faults in woven and knitted textiles.	assists in aligning our research with classical approaches and in identifying common evaluation methods.

Gray Level Co-occurrence Matrix (GLCM) ²	captures the spatial dependence among pixel intensities and is used to describe textures via statistical measures like contrast, correlation, and homogeneity.	feature for describing the surface texture of a material and identifying areas having defects.
Local Binary Patterns (LBP) ³	uses binary codes obtained by comparing each pixel and its neighbouring pixels. It is less sensitive to lighting variations.	potential lightweight descriptor for identifying local anomalies like holes, stains, and broken threads.
Gabor Filters for Texture Analysis ⁴	Uses filters designed for particular frequencies and orientations to determine texture at different scales.	may also aid in detection of weaving direction patterns and defects that are specific to a certain direction.

Challenges Faced

- Understanding the differences between classical texture analysis techniques (GLCM, LBP, Gabor filters), as well as their relevance to fabric images.
- Finding an appropriate open dataset with enough diversity of defects and decent image quality.
- Evaluating texture descriptors from books, without implementing or testing them previously.

Plan for Next Week

- Selection of the final dataset for the project.
- Implementation of prior image processing (resizing, filtering, contrast normalisation).
- Implementation and evaluation of one or two texture descriptors on example images.
- Definition of the pipeline for defect detection in subsequent weeks.

Conclusion

Thus, the project managed to move from its general concept to a defined topic with specified scope and workflow in its first week. Most of the time during this week was spent on background research review and understanding of the classical texture descriptors as well as on repository setup and dataset search. With the midterm examinations coming close, the progress was consciously maintained at the level of planning and researching.

Citations

1. [R. F. A. Silvestre, R. Carrilho, K. A. Hambarde, et al., "Toward Automated Fabric Defect Detection: A Survey of Recent Computer Vision Approaches," *Electronics*, vol. 13, no. 18, p. 3728, Sep. 2024](#)
2. [V. Asha, "Automatic Fabric Inspection using GLCM-based Jensen-Shannon Divergence on Dot-Patterned Fabric," *Informatica*, vol. 46, no. 1, pp. 19–25, 2022](#)

3. [E. Tajeripour, E. Kabir, and A. Sheikhi, "Fabric Defect Detection Using Modified Local Binary Patterns," *EURASIP Journal on Advances in Signal Processing*, vol. 2008, Article ID 353907, 2008](#)
4. [J. Kim, H. Jo, et al., "Automatic fabric defect detection using optimal Gabor filter based on hybrid beetle antennae search–gravitational search algorithm," *Journal of Optics*, vol. 52, no. 4, pp. 1667–1675, 2023](#)